CS471 – Web Technologies (Laboratory)	Qassim University College of Computer	Lab 5
		HTML and CSS (Part 2)

Task 1:

Step 1

Confirmed that `books/` is registered in the URLs and linked to the `bookmodule` app. This ensures that all routes inside `apps/bookmodule/urls.py` are included.

```python
from django.contrib import admin
import apps.bookmodule.views
from django.urls import include, path


urlpatterns = [
    path('admin/', admin.site.urls),
    path('books/', include("apps.bookmodule.urls")), #include urls.py of bookmodule app
    path('users/', include("apps.usermodule.urls")) , #include urls.py of usermodule app
]
```

Step 2

The path `'html5/links/'` is registered to link to `views.links_page`, ensuring that when a user visits `/books/html5/links/`, the correct view is displayed.

```python
from django.urls import path
from . import views
# urlpatterns = [
#     path('', views.index),
#     # path('index2/<int:val1>/', views.inde
#     path('index2/<str:val1>/', views.index2
#     path('<int:bookId>', views.viewbook)
# ]
urlpatterns = [
    # path('', views.index, name= "books.inde
    # path('list_books/', views.list_books, n
    # path('<int:bookId>/', views.viewbook, n
    # path('aboutus/', views.aboutus, name="b
    path('html5/links/', views.links_page),
]
```

Step 3

I created the links_page(request) function in views.py in the book module to display the links.html page

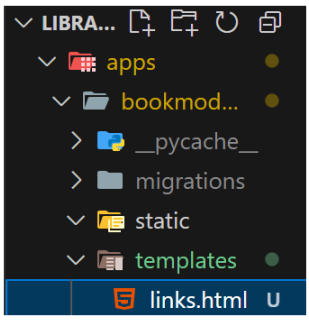
```python
from django.http import HttpResponse
from django.shortcuts import render

def links_page(request):
    return render(request, 'links.html')

```

Step 4

I added this file "link.html" inside the templates of the bookmodule app to create a page displaying the links to QU website. I used **class** for shared styles and



d for unique styling, loaded main.css using {% static 'main.css' %} for styling, and added a JavaScript function showMessage() to show an alert when a link is clicked.

urls.py ...\bookmodule 1, M | urls.py libraryproject 3, M | views.py 2, M | settings.py M | links.html U | main.css U

apps > bookmodule > templates > links.html > ...

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Links Page</title>
    {% load static %}
    <link rel="stylesheet" href="{% static 'main.css' %}">
    <script>
        function showMessage() {
            alert("You clicked a link! 🎉");
        }
    </script>
</head>
<body>
    <div class="link-box" id="box1">
        <a href="https://myqu.qu.edu.sa/login" target="_blank" class="link" id="link1" onclick="showMessage()">Go to QU web
    </div>
    <div class="link-box" id="box2">
        <a href="https://myqu.qu.edu.sa/login" class="link" id="link2" onclick="showMessage()">Go to QU website (open in th
    </div>
    <div class="link-box" id="box3">
        <a href="https://myqu.qu.edu.sa/login" class="link" id="link3" onclick="showMessage()">Go to QU website (open in th
    </div>
</body>
</html>
```

Step 5

I created a CSS file to style the links.html page. I used .link-box for shared styling, including padding and text alignment, while assigning unique IDs (#box1, #box2, #box3) to set different background colors. For links, .link applies common styles, and each link has a unique ID (#link1, #link2, #link3) to define its color. also added hover, visited, and active states to enhance interactivity

```css
.link-box {
    padding: 15px;
    text-align: center;
    display: block;
}
#box1 {
    background-color: lightgreen;
}
#box2 {
    background-color: darkgray;
}
#box3 {
    background-color: rgb(0, 0, 0);
}
.link {
    font-size: 15px;
    padding: 10px;
    text-decoration: underline;
    display: inline-block;
}
#link1 {
    color: red;
}

#link2 {
    color: yellow;
}
```

```css

#link3 {
    color: white;
}
.link:hover {
    color: rgb(15, 34, 93);
    opacity: 0.7;
}
.link:visited {
    color: purple;
}
.link:active {
    color: orange;
    font-weight: bold;
}
```

Output :

Task 2:

Step 1

I went to the urls of the bookmodule app and I add the new path

```
views.py 2, M   formatting.html U   links.html U   main.css U   urls.py 1, M
apps > bookmodule > urls.py > ...
from django.urls import path
from . import views
# urlpatterns = [
#     path('', views.index),
#     # path('index2/<int:val1>/', views.index2)
#     path('index2/<str:val1>/', views.index2, name="book_index2"),
#     path('<int:bookId>', views.viewbook)
# ]
urlpatterns = [
    # path('', views.index, name= "books.index"),
    # path('list_books/', views.list_books, name= "books.list_books"),
    # path('<int:bookId>/', views.viewbook, name="books.view_one_book"),
    # path('aboutus/', views.aboutus, name="books.aboutus"),
    path('html5/links/', views.links_page),
    path('html5/text/formatting', views.text_formatting),
]
```

Step 2

I add the text_formatting function the display the formatting page

```
views.py 2, M   formatting.html U   links.html U
apps > bookmodule > views.py > ...
def text_formatting(request):
    return render(request, 'formatting.html')
```

Step 3

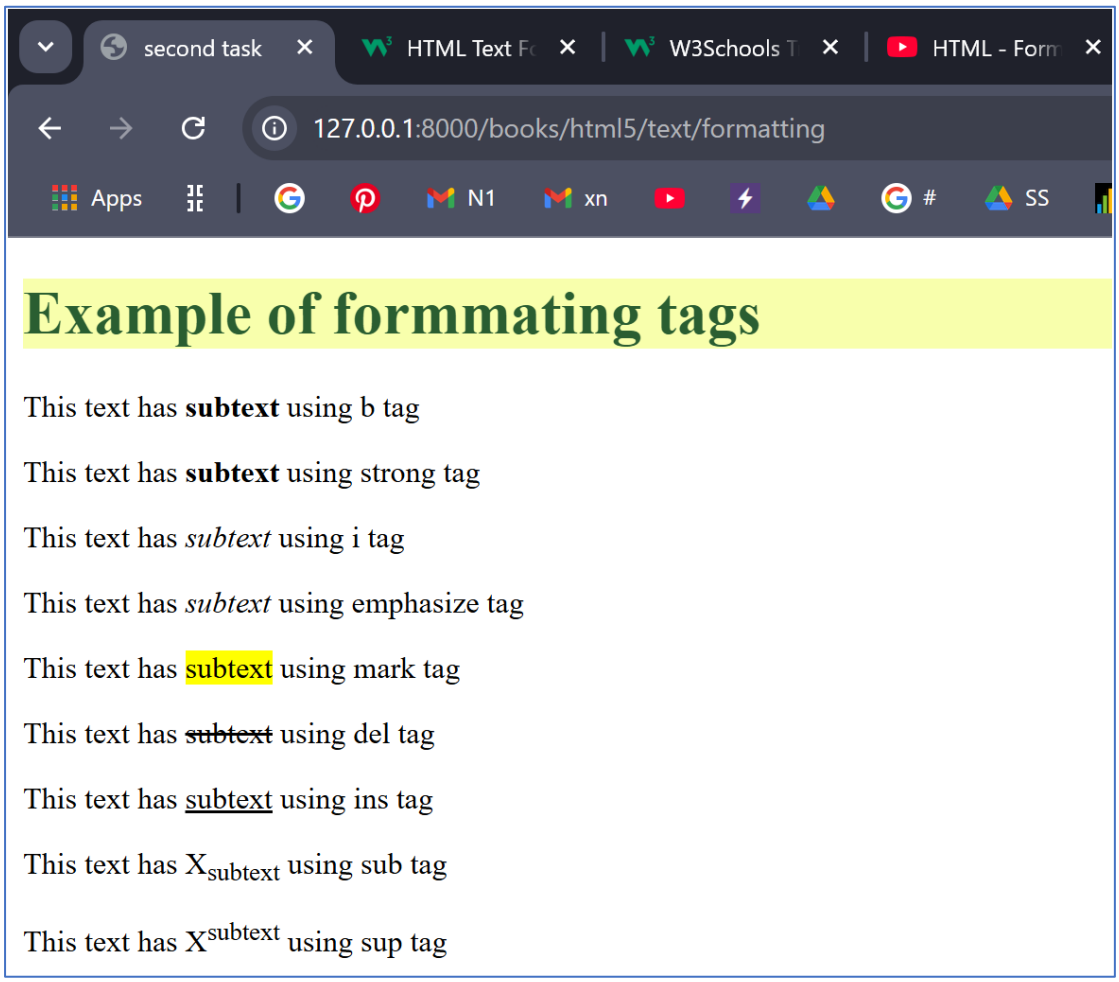
I added the required formatting by going back to w3shcools

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>second task</title>
    {% load static %}
    <link rel="stylesheet" href="{% static 'main2.css' %}">
</head>
<body>
    <h1 class= "first-line">Example of formmating tags</h1>
    <p>This text has <b>subtext</b> using b tag</p>
    <p>This text has <strong>subtext</strong> using strong tag</p>
    <p>This text has <i>subtext</i> using i tag</p>
    <p>This text has <em>subtext</em> using emphasize tag</p>
    <p>This text has <mark>subtext</mark> using mark tag</p>
    <p>This text has <del>subtext</del> using del tag</p>
    <p>This text has <ins>subtext</ins> using ins tag</p>
    <p>This text has X<sub>subtext</sub> using sub tag</p>
    <p>This text has X<sup>subtext</sup> using sup tag</p>
</body>
</html>
```

apps > static > main2.css > ...

```css
.first-line{
    background-color: rgb(248, 255, 172);
    color: rgb(42, 94, 49);
}
```

Output :



Task 3

Step 1

I went to the urls of the bookmodule app and I add the new path

```python
urlpatterns = [
    # path('', views.index, name= "books.index"),
    # path('list_books/', views.list_books, name= "books.list_books"),
    # path('<int:bookId>/', views.viewbook, name="books.view_one_book"),
    # path('aboutus/', views.aboutus, name="books.aboutus"),
    path('html5/links/', views.links_page),
    path('html5/text/formatting', views.text_formatting),
    path('html5/listing', views.listing),
]
```

Step 2

I add the listing function the display the listing page

```
views.py 2, M    listing.html U    main3.css U
apps > bookmodule > views.py > ...
48  def listing(request):
49      return render(request, 'listing.html')
```

Step 3

I created 2 files 1- listing.html in templates 2- main3.css in static

```
LIBRARYPROJECT
  apps
    bookmodule
      templates
        formatting.html   U
        links.html        U
        listing.html      U
      __init__.py
      admin.py
      apps.py
      models.py
      tests.py
      urls.py             1, M
      views.py            2, M
    static
      book1.jpg
      book2.png
      book3.png
      main.css            U
      main2.css           U
      main3.css           U
```

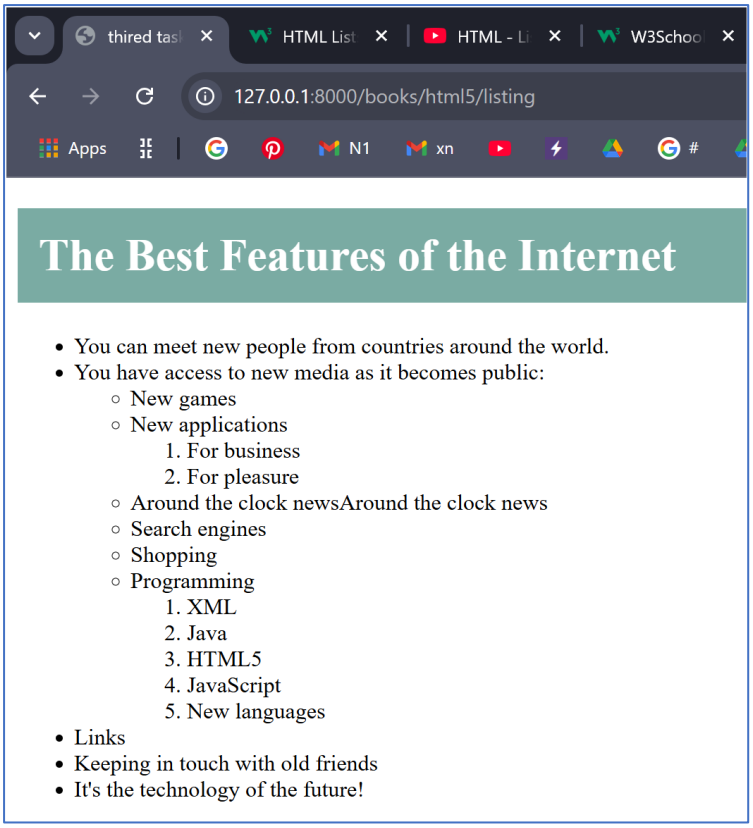
Step 4

I created this file to display a the structured list. I used a <h1> with the class first-line for styling the title and a nested <ul> for listing various features. Different list-style-type values were applied to organize the sublists clearly. The external CSS file main3.css was linked for styling. I used to W3Schools for guidance in structuring and styling the lists

```html
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>thired task</title>
    {% load static %}
    <link rel="stylesheet" href="{% static 'main3.css' %}">
</head>
<body>
    <h1 class= "first-line">The Best Features of the Internet</h1>
    <ul>
        <li>You can meet new people from countries around the world.</li>
        <li>You have access to new media as it becomes public:
            <ul style="list-style-type: circle;">
                <li>New games</li>
                <li>New applications
                    <ul style="list-style-type:decimal;">
                        <li>For business</li>
                        <li>For pleasure</li>
                    </ul>
                </li>
                <li>Around the clock newsAround the clock news</li>
                <li>Search engines</li>
                <li>Shopping</li>
                <li>Programming
                    <ul style="list-style-type:decimal;">
                        <li>XML</li>
                        <li>Java</li>
                        <li>HTML5</li>
                        <li>JavaScript</li>
                        <li>New languages</li>
                    </ul>
                </li>
            </ul>
        </li>
        <li>Links</li>
        <li>Keeping in touch with old friends</li>
        <li>It's the technology of the future!</li>
    </ul>
</body>
```

```
views.py 2, M    listing.html U    main3.css U
apps > static > main3.css > ...
.first-line{
    background-color: rgb(122, 171, 163);
    color: rgb(255, 255, 255);
    padding: 15px 15px;
}
```

Output :



Task 4

first table (#fruits-table):

I applied a border to the entire table and its cells using border: 1px solid black , I ensured text alignment is centered using text-align: center , To center the table on the page, I used margin: auto , I added a caption with a heading, aligning it to the center for a clear title.

```
#fruits-table, #fruits-table th, #fruits-table td {
    border: 1px solid black;
    text-align: center;
}
```

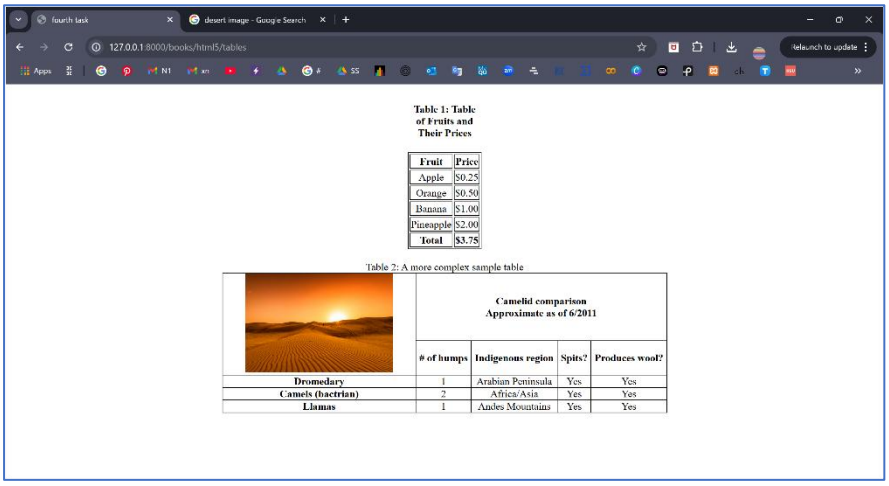
```
<body>
    <table id="fruits-table" style="margin: auto; text-align: center;">
        <caption>
            <h4 style="text-align: center;">
                Table 1: Table<br>of Fruits and<br>Their Prices
            </h4>
        </caption>
        <tr>
            <th>Fruit</th>
            <th>Price</th>
        </tr>
        <tr>
            <td>Apple</td>
            <td>$0.25</td>
        </tr>
        <tr>
            <td>Orange</td>
            <td>$0.50</td>
        </tr>
        <tr>
            <td>Banana</td>
            <td>$1.00</td>
        </tr>
        <tr>
            <td>Pineapple</td>
            <td>$2.00</td>
        </tr>
        <tr>
            <th>Total</th>
            <th>$3.75</th>
        </tr>
    </table>
```

second table (#second-table):

1. **Added a border and collapsed it:** I made sure the table and its cells have clear borders by using border: 1px solid black;. I also applied border-collapse: collapse; to remove extra spacing between borders and make the table look neater.
2. **Centered the table and text:** I used margin: auto; to center the table on the page and text-align: center; to ensure all text is properly aligned in each cell.
3. **Inserted an image inside the first cell:** The desert image is placed in the first column using <img src="{% static 'desert.jpg' %}">. Since this image spans two rows (rowspan="2"), I made sure it stays aligned. To add some space around it, I used margin-left: 30px; margin-right: 30px.
4. **Merged cells for the title row:** The second cell in the first row spans across four columns (colspan="4") to display the title "Camelid comparison" properly without being broken into separate cells.
5. **Added padding for better spacing**: I used padding: 0px 5px; inside the cells so that text doesn't touch the cell borders too closely.

```html
<table id="second-table" style="margin: auto; text-align: center;" >
    <caption>
        <label>Table 2: A more complex sample table</label>
    </caption>
    <tr>
        <td style="text-align: center;" rowspan="2">
            <img src="{% static 'desert.jpg' %}" alt="Desert Image"
            width="230" style="margin-left: 30px; margin-right: 30px;">
            <td colspan="4"><b>Camelid comparison<br>Approximate as of 6/2011</b></td>
    </tr>
    <tr>
        <th># of humps</th>
        <th>Indigenous region</th>
        <th>Spits?</th>
        <th>Produces wool?</th>
    </tr>
    <tr>
        <th>Dromedary</th>
        <td>1</td>
        <td>Arabian Peninsula</td>
        <td>Yes</td>
        <td>Yes</td>
    </tr>
    <tr>
        <th>Camels (bactrian)</th>
        <td>2</td>
        <td>Africa/Asia</td>
        <td>Yes</td>
        <td>Yes</td>
    </tr>
    <tr>
        <th>Llamas</th>
        <td>1</td>
        <td>Andes Mountains</td>
        <td>Yes</td>
        <td>Yes</td>
    </tr>
```

```html
{% load static %}
```

```css
#second-table , #second-table th, #second-table td {
    border: 1px solid black;
    text-align: center;
    border-collapse: collapse;
}
#second-table th, #second-table td {
        padding:  0px 5px;
        text-align: center;
    }
```

Output :